

Shale gas resources are found in numerous basins with organic-rich shales. In India, four priority basins are Cambay, Krishna-Godavari, Cauvery and Damodar Valley. Thus all the given codes are correct and (d) is the correct answer.

38. Which is not a fossil fuel?

- (a) Coal (b) Wood
(c) Diesel (d) Petrol
(e) None of these

Chhattisgarh P.C.S. (Pre) 2014

Ans. (b)

The remains of plants and animals buried under the Earth, millions of years ago are called fossils. These fossils are excellent fuels called fossil fuels. The examples of fossil fuels are coal, petroleum, diesel, kerosene and natural gas. Wood is not a fossil fuel; but it is a biofuel.

39. Choose the correct option representing the following pipelines Transports in decreasing order of their lengths?

1. Salaya-Koyali-Mathura pipeline
 2. Hajira-Bijapur-Jagdishpur gas pipeline
 3. Jamnagar-Loni (LPG) pipeline
 4. Kandala-Bhatrinda pipeline
- (a) $1 > 2 > 3 > 4$ (b) $2 > 4 > 1 > 3$
(c) $2 > 4 > 3 > 1$ (d) $2 > 3 > 4 > 1$

Chhattisgarh P.C.S. (Pre) 2020

Ans. (*)

The length of the Salaya-Koyali-Mathura pipeline is 2663 km (as of April 1, 2024). It starts from Salaya near Vadinar in Jamnagar district of Gujarat and supplies crude oil to Indian Oil's refineries at Koyali in Gujarat, Mathura in Uttar Pradesh and Panipat in Haryana. The length of the Hazira - Bijapur (Vijaipur) - Jagdishpur gas pipeline is 6718 km. (as of June, 2024). It is a GAIL project. This pipeline passes through Uttar Pradesh, Madhya Pradesh, Rajasthan and Gujarat, Haryana, Delhi and Uttarakhand.

The Jamnagar - Loni (LPG) pipeline starts from Jamnagar in Gujarat and goes to Loni near Ghaziabad in Uttar Pradesh. Its length is 1427 km (as of December 31, 2024). The length of the Kandla - Bhatrinda pipeline is 1413 km (as of April 1, 2024). Now it is known as the Panipat-Bhatrinda pipeline (219 km.) and Mundra-Panipat (1194 km.) pipeline. Chhattisgarh Public Service Commission has issued answer option (c) for this question, which is incorrect.

E. Miscellaneous : Minerals

*Minerals are generally found in the earth's crust in ore form. Several types of the minerals are found in a different regions of India, but these are unequally distributed. *Earlier, India was first in mica production. Mica was declared a minor mineral on 10 Feb, 2015 by the Indian Government. According to the Indian Minerals Yearbook 2022, the minor minerals are barytes, Bentonite, Calcite, Corundum (Minor) & Sapphire (Major), Diaspore, Dolomite, Dunite & Pyroxenite, Feldspar, Fireclay, Fuller's Earth, Granite, Gypsum (Minor) & Selenite (Major), Kaolin, Ball Clay, Other Clay & Shale, Laterite, Marble, Mica, Ochre, Other Calcareous Materials, Pyrophyllite, Quartz & Other Silica Minerals [Moulding Sand and Flint Stone (Major)], Slate, Sandstone & Other Dimension Stones, Talc, Soapstone and Steatite.

*The only producer of **asbestos** as per the data of the year 2013-14 is **Andhra Pradesh**, whereas its largest reserves/resources (as of April 1, 2020, P) are found in **Rajasthan**. No production of asbestos was reported in 2021-22 as well as in the previous year and no reporting mines were noted in 2021-22 as well as in the preceding year. *Metamorphic rock is a result of the transformation of existing rock, in a process called metamorphism. *Metamorphic minerals are formed under high temperature and pressure. **Garnet** is associated with metamorphism.

Five leading states in manganese ore production as per the figures of the year, 2021-22 (P) are-

1. Madhya Pradesh 849221 tonnes
2. Maharashtra 731730 tonnes
3. Odisha 512591 tonnes
4. Karnataka 380004 tonnes
5. Andhra Pradesh 204002 tonnes

***Manganese** is mainly found in **Balaghat** and **Chhindwara** districts of Madhya Pradesh and, Bhandara and Nagpur districts of Maharashtra. Manganese extracted from Odisha is rich in iron content and lacks in phosphorus. **Balaghat** is an important district of Madhya Pradesh in the perspective of mineral resources. The major minerals found here are **kyanite, manganese ore, copper** etc. *Nickel is a lustrous, silvery-white metal. It is the fifth most common element in the Earth's crust. Nickel does not occur in a native state. The **largest resources** (as of April 1, 2020, P) of **nickel** in India are present in Odisha, but it is not mined. The entire demand for nickel in India is met through imports.

***Lead** mainly occurs in metamorphic rock (schist) in a combined form with silver. It is also found in Precambrian and Vindhyan limestone. Lead is resistant with respect to acidic matter and atmospheric corrosion. *According to

data from the Indian Minerals Yearbook, 2022, **Rajasthan** is the only producer of lead in India. The **Zawar region** is famous for lead production. ***Gypsum** is found in large quantities in Churu-Bikaner-Sriganganagar belt. It is used for improving soil fertility. It is also used in the health and construction sectors after value addition. *According to the Indian Minerals Yearbook 2022, **Rajasthan** is the largest (2021-22) producer of Gypsum.

***Jaduguda Mine** in eastern Singhbhum district of Jharkhand is famous for **uranium production**. Other uranium production regions are **Domiasiat** (Meghalaya), **Lambapur** (Andhra Pradesh), **Rohail** (Rajasthan) and **Gogi** (Karnataka). In Uttar Pradesh, uranium is found in **Lalitpur**. **Monazite** is an important ore of thorium, lanthanum, uranium and cerium. It is found in large quantities in the coastal areas of **Kerala**. *Large reserves of diamond bearing **kimberlite** tracts have been discovered in Payalikhanda and Behradin in Raipur district of Chhattisgarh.

Mineral Production in India (2021-22, P)			
Minerals	First	Second	Third
Bauxite	Odisha	Gujarat	Jharkhand
Chromite	Odisha	-----	-----
Coal	Odisha	Chhattisgarh	Madhya Pradesh
Graphite	Odisha	Tamil Nadu	Jharkhand
Limestone	Rajasthan	Andhra Pradesh	M.P.
Diamond	M.P.	-----	-----
Manganese Ore	M.P.	Maharashtra	Odisha
Crude oil*	Rajasthan	Gujarat	Assam
Natural Gas*	Assam	Rajasthan	Tripura
Copper Ore	M.P.	Rajasthan	Jharkhand
Iron Ore	Odisha	Chhattisgarh	Karnataka
Gold Ore	Karnataka	Jharkhand	Andhra Pradesh
Silver	Karnataka	-----	-----
Note : * 2020-21 (P)			

*According to data from the Indian Minerals Yearbook-2022, **Odisha** continues to be the major **chromite producing state** accounting for almost the entire production during 2021-22 (P). **Kolar** and **Raichur** districts are famous for gold production in Karnataka. Gold is also extracted from Hutti Gold Mine (Raichur). ***Panna**, located in Madhya Pradesh, is famous for its diamond mines.

1. Consider the following minerals :

1. Bentonite
2. Chromite

3. Kyanite

4. Sillimanite

In India, which of the above is/are officially designated as major minerals?

- (a) 1 and 2 only
- (b) 4 only
- (c) 1 and 3 only
- (d) 2, 3 and 4 only

I.A.S. (Pre) 2020

Ans. (d)

Indian Minerals Yearbook 2022, Bentonite has been designated as a minor mineral. Other minor minerals are : fuller's earth, marble, mica etc.

2. Which one of the following statements is true?

- (a) Odisha is the largest producer of Chromite in India.
- (b) Karnataka is the largest producer of Silver in India.
- (c) Andhra Pradesh has the largest reserve of Bauxite in India
- (d) Odisha has the largest reserve of Manganese in India

U.P. B.E.O. (Pre) 2019

Ans. (*)

According to figure of Indian Minerals Yearbook 2022, Odisha has a monopoly of Chromite production. Karnataka is the top silver producer (2021-22, P) in India. Odisha has the largest (April 1, 2020, P) reserves/resources of Manganese Ore in India. Odisha has the largest reserves/resources (April 1, 2020, P) of Bauxite in India.

3. Which of the following are NOT correctly matched?

(Mineral)	(Mining City)
1. Copper	- Chitradurga
2. Iron ore	- Ballari
3. Manganese	- Bhilwara
4. Bauxite	- Katni

Select the correct answer from the codes given below:

Codes :

- (a) 1 and 2
- (b) 2 and 3
- (c) 1 and 3
- (d) 2 and 4

U.P. P.C.S. (Pre) 2020

Ans. (c)

Bellari, located in Karnataka, is famous for iron ore mining. Iron ore and manganese are mined in Chitradurga. Bhilwara (Rajasthan) is famous for copper mining. Manganese ore deposits in Rajasthan are found in Banswara, Udaipur and Pali; Katni is known for mining of Bauxite in Madhya Pradesh.

4. Which of the following states is the largest producer of asbestos in India?

- (a) Madhya Pradesh
- (b) Bihar

(c) Karnataka

(d) Rajasthan

U.P. Lower Sub. (Pre) 2015

Ans. (*)

According to 2013-14 data, the only producer of asbestos in India is Andhra Pradesh. According to the Indian Minerals Yearbook 2022, the largest reserve of asbestos is in Rajasthan (as of April 1, 2020, P). As the question is related to production, thus, no answer is correct.

5. Consider the following statement :

1. Natural gas occurs in the Gondwana belts.

2. Mica occurs in abundance in Kodarma.

3. Dharwars are famous for petroleum.

Which of the statements given above is/are correct?

(a) 1 and 2

(b) Only 2

(c) 1 and 3

(d) None of these

I.A.S. (Pre) 2013

Ans. (b)

Petroleum and natural gas are found in the sedimentary rocks of the tertiary plate. Gondwana belts belong to the upper Palaeozoic and Mesozoic periods and the Dharwar rocks belong to the late Archean period. Both these rocks are of a much older period and consist mostly of metamorphosed rocks and thus are very poor in petroleum and natural gases. Hence, both statements 1 and 3 are incorrect. Kodarma lies in Jharkhand and is quite famous for its mica. That is why it is called as the 'Mica City of India'. Thus, only statement 2 is correct and the right answer is option (b).

6. Which of the following statements is correct?

(a) Natural gas is found in Dharwar rock formation.

(b) Mica is found in Kodarma.

(c) Cuddapah series is famous for diamonds.

(d) Petroleum reserves are found in Aravalli hills.

69th B.P.S.C. (Pre), 2023

Ans. (b)

Natural gas is found in sedimentary basins, generally in the areas of marine sediments. Hence statement (a) is incorrect. Significantly, Kodarma district of Jharkhand state has been famous worldwide for mica mining, especially for ruby mica. The main reserve of mica is found under the forest of Wildlife Sanctuary of Kodarma. Cuddapah basin has been well known for- barytes, asbestos, copper, iron, lead, diamond, uranium etc. Petroleum is found in Krishna, Godavari, Cauvery basin, Assam, Gujarat and Rajasthan.

7. Which among the following is connected with Metamorphic rock ?

(a) Copper

(b) Garnet

(c) Manganese

(d) Pyrite

U.P. Lower Sub. (Spl.) 2010

Ans. (b)

Metamorphic rocks are a result of a transformation of existing rock, in a process called metamorphism, which means "change in form". Metamorphic minerals are those that form only at the high temperatures and pressures associated with the process of metamorphism.

8. Quartzite is metamorphosed from :

(a) Limestone

(b) Obsidian

(c) Sandstone

(d) Shale

I.A.S. (Pre) 2001

Ans. (c)

Quartzite is a non-foliated metamorphic rock composed almost entirely of quartz. It forms when a quartz-rich sandstone is altered by the heat, pressure and chemical activity of metamorphism.

9. Match List-I (Ores) with List-II (State where they are mined) and select the correct answer using the codes given below the lists:

List- I

List- II

A. Manganese

1. Madhya Pradesh

B. Nickel

2. Odisha

C. Lead-zinc

3. Rajasthan

D. Asbestos

4. Andhra Pradesh

Code :

	A	B	C	D
(a)	1	3	2	4
(b)	4	3	2	1
(c)	1	2	3	4
(d)	4	2	3	1

U.P.P.C.S. (Pre) 1998

I.A.S. (Pre) 1996

Ans. (c)

List I	List II
Manganese	Madhya Pradesh
Nickel	Odisha
Lead-zinc	Rajasthan
Asbestos	Andhra Pradesh

10. The leading producer of Manganese in India is –

- (a) Karnataka (b) Maharashtra
(c) Madhya Pradesh (d) Odisha

U.P.P.C.S. (Pre) 2012
U.P.P.C.S. (Mains) 2008

Ans. (c)

According to data of Indian Minerals Yearbook 2022, largest producer of Manganese Ore state is Madhya Pradesh (2021-22,P).

11. Which is the correct sequence of following manganese producing states of India in terms of high to low production status in the year 2013-14?

- (a) Rajasthan, Odisha, Madhya Pradesh and Andhra Pradesh
(b) Odisha, Madhya Pradesh, Andhra Pradesh and Rajasthan
(c) Madhya Pradesh, Odisha, Andhra Pradesh and Rajasthan
(d) Madhya Pradesh, Odisha, Rajasthan and Andhra Pradesh

R.A.S./R.T.S. (Pre) 2016

Ans. (c)

During the time, when this question was asked, option (c) was the correct answer. Presently the correct sequence of following Manganese Ore producing states of India in terms of high to low in the year 2021-22 (P) is, Madhya Pradesh > Maharashtra > Odisha > Karnataka > Andhra Pradesh.

12. Match List-I with List-II and select the correct answer by using the codes given below the lists:

List- I (Mineral)	List- II (Top producing states)
A. Iron Ore	1. Odisha
B. Copper	2. Karnataka
C. Gold	3. Rajasthan
D. Mica	4. Andhra Pradesh

Code :

	A	B	C	D
(a)	1	3	2	4
(b)	4	2	3	1
(c)	1	4	2	3
(d)	3	1	4	2

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (*)

(Minerals)	(Top producing States)
Iron ore	– Odisha
Copper Ore	– Madhya Pradesh
Gold Ore	– Karnataka
Mica	– Rajasthan

13. India has attained self-sufficiency in the production of which mineral?

- (a) Tin (b) Silver
(c) Gold (d) Graphite

U.P.P.C.S. (Pre) 1991

Ans. (*)

India has not attained self-sufficiency in the production of any mineral mentioned in the question. According to data of Indian Minerals Yearbook 2022, world reserves of Graphite have been placed at 320 million tonnes of which Turkey accounts for 28%, followed by China (23%), Brazil (22%), Madagascar & Mozambique (8% each) and Tanzania 5%, India & Uzbekistan (2% each).

14. Match List-I with List-II and select the correct answer using the codes given below the lists:

List- I	List- II
A. Mineral Oil	1. Odisha
B. Gypsum	2. Karnataka
C. Gold	3. Gujarat
D. Bauxite	4. Rajasthan

Code :

	A	B	C	D
(a)	3	2	1	4
(b)	2	1	4	3
(c)	4	3	1	2
(d)	3	4	2	1

U.P. Lower Sub. (Spl.) (Pre) 2010
U.P.U.D.A./L.D.A. (Pre) 2006

Ans. (*)

According to (Indian Minerals Yearbook 2022) given options, the correct match is as below:

Minerals	States
Mineral Oil (Cru)	– Rajasthan
Gypsum	– Rajasthan
Gold	– Karnataka
Bauxite	– Odisha

15. Which state of India is renowned for its deposits of limestone, gypsum, salt and zinc?

- (a) Haryana (b) Gujarat
(c) Punjab (d) Rajasthan

U.P. R.O./A.R.O. (Pre) 2023

Ans. (d)

Rajasthan state in India is famous for its deposits of limestone, gypsum, salt and zinc.

16. Which of the following is not a metallic mineral?

- (a) Haematite (b) Bauxite
(c) Gypsum (d) Limonite

Jharkhand P.C.S. (Pre) 2011

Ans. (c)

India has various non-metallic mineral reserves. Non-metallic minerals available from India are Phosphorite, Dolomite, Gypsum, Garnet, Wollastonite, Vermiculite, Ochre, Perlite, Bentonite, Asbestos, Cadmium, Feldspar, Soapstone, Kaolin, Sillimanite, Limestone, Diatomite, Pyrophyllite, Fluorite, Vanadium, Dunite, Ilmenite, Gallium and Zircon.

17. The main gypsum producing state of India is

- (a) Bihar (b) Jharkhand
(c) Chhattisgarh (d) Karnataka
(e) Rajasthan

Chhattisgarh P.C.S. (Pre) 2013

Ans. (e)

According to the Indian Minerals Yearbook - 2022, Rajasthan is the leading producer of gypsum in India during 2021-22 (P).

18. What is the product found in large quantities in Churu-Bikaner-Sri Ganganagar belt, which is (i) a source of environmental pollution, (ii) used for raising soil fertility, and (iii) after value addition, is used in health and building sector?

- (a) Limestone (b) Lignite
(c) Fullers Earth (d) Gypsum

R.A.S./R.T.S. (Pre) 2008

Ans. (d)

Gypsum is found in large quantities in the Churu-Bikaner-Sri Ganganagar belt. It is a source of environmental pollution. It is used for raising soil fertility and after value addition, it is used in the health and construction sectors.

19. Match List-I with List-II and select the correct answer using the codes given below the lists:

- | List- I | List- II |
|--------------|--------------|
| A. Coal | 1. Bhandara |
| B. Gold | 2. Karanpura |
| C. Mica | 3. Hutti |
| D. Manganese | 4. Nellore |

Codes :

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 1 | 3 | 2 | 4 |
| (b) | 2 | 3 | 4 | 1 |
| (c) | 3 | 4 | 2 | 1 |
| (d) | 2 | 1 | 4 | 3 |

I.A.S. (Pre) 1997

Ans. (b)

Coal is found in Karanpura in Jharkhand, Gold is found in the mines of Hutti in Karnataka, Mica is found in the Nellore district of Andhra Pradesh and Manganese is found in Bhandara mines in Maharashtra.

20. Match List-I with List-II and select the correct answer from the codes given below :

- | List -I
(Centre) | List-II
(Mineral) |
|---------------------|----------------------|
| A. Makum | 1. Iron Ore |
| B. Dallirajhara | 2. Coal |
| C. Koraput | 3. Manganese |
| D. Chitradurg | 4. Bauxite |
- Code :

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 3 | 2 | 1 | 4 |
| (b) | 2 | 1 | 4 | 3 |
| (c) | 4 | 3 | 2 | 1 |
| (d) | 1 | 2 | 3 | 4 |

U.P.P.C.S. (Pre) 2016

Ans. (b)

The correct match of Minerals with their centres is as follows:

- | (Centre) | (Minerals) |
|--------------|-------------|
| Makum | - Coal |
| Dallirajhara | - Iron Ore |
| Koraput | - Bauxite |
| Chitradurg | - Manganese |

21. Which of the following is not correctly matched?

- | | |
|---------------|-------------|
| (a) Amguri | - Petroleum |
| (b) Lanjigarh | - Bauxite |
| (c) Kamptee | - Copper |
| (d) Bellary | - Iron ore |

U.P.P.C.S. (Mains) 2009

Ans. (c)

According to given options, correct matching is given below:

- | (Place) | (Mineral) |
|-----------------------|-------------|
| Amguri (Assam) | - Petroleum |
| Lanjigarh (Odisha) | - Bauxite |
| Kamptee (Maharashtra) | - Coal |
| Bellary (Karnataka) | - Iron ore |
- Thus, option (c) is not correctly matched.

22. Match List-I with List-II and select the correct answer using the codes given below the list:

List- I (Region)	List- II (Mineral)
A. Badam Pahar	1. Copper
B. Kodarma	2. Iron ore
C. Mosabani	3. Petroleum
D. Rava	4. Mica

Code :

	A	B	C	D
(a)	4	3	1	2
(b)	2	4	3	1
(c)	1	2	4	3
(d)	2	4	1	3

U.P.P.C.S. (Pre) 2008

Ans. (d)

According to given options, correct match is given below :

Region	Mineral
Badam Pahar (Odisha)	- Iron ore
Kodarma (Jharkhand)	- Mica
Mosabani (Jharkhand)	- Copper
Rava (Krishna-Godavari Basin)	- Petroleum

23. Match List-I with List-II and select the correct answer using the codes given below the lists:

List- I (Iron Ore zone)	List- II (State)
A. Badam Pahar	1. Jharkhand
B. Dalli-Rajhara	2. Odisha
C. Kudremukh	3. Chhattisgarh
D. Noamundi	4. Karnataka

Code :

	A	B	C	D
(a)	4	2	3	1
(b)	1	3	4	1
(c)	2	3	4	1
(d)	2	1	4	3

U.P.U.D.A./L.D.A. (Mains) 2010

Ans. (c)

According to given options, correct match is given below :

Iron ore zone	State
Badam Pahar	- Odisha
Dalli-Rajhara	- Chhattisgarh
Kudremukh	- Karnataka
Noamundi	- Jharkhand

24. Which one of the following is not correctly matched?

(a) Dalli-Rajhara - Iron Ore

- (b) Rakha - Copper
(c) Nellore - Manganese
(d) Amar Kantak - Bauxite

U.P.P.C.S. (Mains) 2002

Ans. (c)

Nellore, located in Andhra Pradesh, is famous for the production of mica and copper not for manganese. Thus, option (c) is not correctly matched.

25. Match List-I with List-II and select the correct answer using the codes given below the lists :

List- I (Minerals)	List- II (Mining area)
A. Graphite	1. Bellary
B. Lead	2. Didwana
C. Salt	3. Rampa
D. Silver	4. Zawar

Code :

	A	B	C	D
(a)	3	4	2	1
(b)	1	4	2	3
(c)	3	1	4	2
(d)	2	3	1	4

I.A.S. (Pre) 1998

Ans. (a)

The correct match of minerals and their respective mining areas is mentioned below :

Minerals	Mining Areas
Graphite	- Rampa (Andhra Pradesh)
Lead	- Zawar (Rajasthan)
Salt	- Didwana (Rajasthan)
Silver	- Bellary (Karnataka)

26. Match List-I with List-II and select the correct answer using the codes given below the lists :

List- I (Place)	List- II (Minerals)
A. Ankleshwar	1. Iron ore
B. Dalli-Rajhara	2. Petroleum
C. Kodarma	3. Copper
D. Khetri	4. Mica

Codes :

	A	B	C	D
(a)	3	4	2	1
(b)	2	1	4	3
(c)	3	1	4	2
(d)	2	3	1	4

U.P.P.C.S. (Pre) 2001

Ans. (b)

Ankleshwar oilfield is located in the Bharuch district of the State of Gujarat.

Dalli Rajhara is home to iron ore captive mines for Bhilai Steel Plant, the largest integrated steel plant in India.

Kodarma district of Jharkhand is known for mica. Copper is largely obtained from Khetri- Singhana, Kho-Dariba and Bhan- goni mines.

Thus, option (b) is correctly matched.

27. Match List-I with List-II and select the correct answer using the codes given below the lists:

List- I (Minerals)	List- II (Areas of Production)
A. Iron Ore	1. Khetri
B. Copper	2. Keonjhar
C. Coal	3. Kodarma
D. Mica	4. Korba

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	4	3	2	1
(c)	2	1	4	3
(d)	3	2	4	1

U.P.P.C.S. (Mains) 2005

Ans. (c)

According to given options, correct match is as follows :

Minerals		Production zone
Iron ore	-	Keonjhar
Copper	-	Khetri
Coal	-	Korba
Mica	-	Kodarma

28. Find the correct sequence of Copper, Gold, Iron, Coal in the following :

- Khetri- Kolar-Kudremukh-Jharia
- Kolar-Khetri-Kudremukh-Jharia
- Jharia-Kolar-Kudremukh-Khetri
- Khetri-Kudremukh-Kolar-Jharia

U.P.U.D.A./L.D.A. (Pre) 2001

Ans. (a)

Khetri mines of Rajasthan are a major copper producing mine of the country, Kolar mines of Karnataka are a major gold producing mine of the country. The Kudremukh mine is one of the largest iron ore mines in the world. Jharia is a notified area in Dhanbad district in Jharkhand State. Jharia is famous for its rich coal resources, used to make coke. Thus, option (a) is the correct sequence of Copper, Gold, Iron and Coal in India.

29. Match the following and select the correct answer by using code given below :

List- I	List- II
A. Gold	1. Khetri
B. Coal	2. Kolar
C. Copper	3. Kudremukh
D. Iron	4. Jharia

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	4	3	2	1
(c)	3	4	1	2
(d)	2	4	1	3

U.P.P.C.S. (Pre) 2008

Ans. (d)

See the explanation of the above question.

30. Which of the following states has a monopoly in chromite production?

- Bihar
- Karnataka
- Maharashtra
- Odisha

U.P.P.C.S. (Spl) (Mains) 2008

Ans. (d)

According to data published by the Indian Bureau of Mines in Indian Minerals Yearbook - 2022, Odisha continues to be the major chromite producing (2021-22, P) state, accounting for almost the entire production.

31. Match the following :

Minerals	Area
A. Gypsum	1. Jhamar-kotra
B. Copper	2. Rampura-agucha
C. Rock phosphate	3. Khoh-Dariba
D. Lead & Zinc	4. Jamsar

Code :

	A	B	C	D
(a)	3	2	4	1
(b)	2	3	4	1
(c)	4	3	1	2
(d)	1	4	2	3

R.A.S./R.T.S. (Pre) 1997

Ans. (c)

According to given options, correct matching is as follows :

Minerals Reserve	Area
Gypsum	— Jamsar
Copper	— Khoh-Dariba
Rock phosphate	— Jhamar-kotra
Lead & Zinc	— Rampura-agucha

32. Match the following :

- | | |
|---------------|-----------|
| (A) Khetri | 1. Iron |
| (B) Jharia | 2. Copper |
| (C) Singhbhum | 3. Coal |

Code :

- (a) 1C, 2A, 3B
 (b) 1A, 2C, 3B
 (c) 1C, 2B, 3A
 (d) 1A, 2B, 3C

M.P.P.C.S. (Pre) 1991

Ans. (a)

Khetri has copper reserves in Jhunjhunu district of Rajasthan. Jharia, located in Jharkhand, is famous for its rich coal resources used to make coke. Singhbhum of Jharkhand is known for its iron ore deposits. Hence, (a) is the correct answer.

33. Which one of the following is not correctly matched ?

- (a) Bailadila-Madhya Pradesh
 (b) Kemanagundi-Karnataka
 (c) Singhbhum-Jharkhand
 (d) Mayurbhanj - Odisha

U.P.P.C.S. (Mains) 2004

Ans. (a)

According to given options, the correct match of mineral production & concerned State is as follows :

Mineral production	State
Bailadila	— Chhattisgarh
Kemanagundi	— Karnataka
Singhbhum	— Jharkhand
Mayurbhanj	— Odisha

34. Match the following mineral and related district –

- | | |
|--------------|--|
| (A) Diamond | (1) Bastar, Durg |
| (B) Iron ore | (2) Panna |
| (C) Bauxite | (3) Surguja, Mandla, Satna
Balaghat, Bilaspur |
| (D) Coal | (4) Sidhi, Surguja, Bilaspur
Raigarh, Shahdol, Chhindwara,
Betul |

Code :

- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 2 | 1 | 3 | 4 |
| (b) | 1 | 4 | 3 | 2 |
| (c) | 3 | 4 | 2 | 1 |
| (d) | 4 | 2 | 3 | 1 |

M.P.P.C.S. (Pre) 1994

Ans. (a)

Panna, located in Madhya Pradesh, is famous for its diamond mines. Iron ore deposits of Chhattisgarh are associated with the rocks that are known to contain high-grade iron ores (Haematite). Major iron ore regions are Rajnandgaon, Durg, Dantewada, Bastar & Bilaspur. High-grade bauxite ore is found in the plateau region of Surguja, Durg, Raigarh and Bilaspur districts in Chhattisgarh. Coal deposits are found in Sidhi, Shahdol, Chhindwada and Betul districts in Madhya Pradesh.

35. Match List-I with List-II and select the correct answer using the codes given below the lists:

- | List- I | List- II |
|-------------|--------------|
| A. Coal | 1. Kudremukh |
| B. Copper | 2. Zawar |
| C. Iron Ore | 3. Khetri |
| D. Silver | 4. Talcher |

Code :

- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 1 | 2 | 3 | 4 |
| (b) | 2 | 1 | 4 | 3 |
| (c) | 4 | 3 | 1 | 2 |
| (d) | 3 | 2 | 4 | 1 |

U.P.P.C.S. (Pre) 1998

Ans. (c)

The Kudremukh mine is one of the largest iron ore mines in the world. The mining takes place in the Western Ghats at Kudremukh located in Chikmagalur district of Karnataka. Zawar mines of Udaipur (Rajasthan) are the largest producers of silver. Khetri - Singhana mines, located in Jhunjhunu district of Rajasthan, are known for copper mining. Talcher Coalfield is located in Angul district of Odisha, on the bank of the Brahmani river.

36. Which one of the following is not correctly matched?

- | | |
|---------------|-------------|
| (a) Iron Ore | — Kudremukh |
| (b) Manganese | — Koraput |
| (c) Copper | — Khetri |
| (d) Coal | — Singareni |

U.P.R.O./A.R.O. (Mains) 2013

Ans. (b)

Koraput is famous for aluminium not for manganese. Other options are correctly matched.

37. Match List-I with List-II and select the correct answer using the codes given below the lists:

List- I

A. Copper

B. Gas plant

C. Aluminium

D. Petroleum

Code :

	A	B	C	D
(a)	4	2	3	1
(b)	1	2	3	4
(c)	4	3	2	1
(d)	3	4	1	2

List- II

1. Kochi

2. Auraiya

3. Korba

4. Khetri

U.P.P.C.S. (Mains) 2002

Ans. (a)

According to the given options, the correct match is as follows :

Copper	—	Khetri
Gas Plant	—	Auraiya
Aluminium	—	Korba
Petroleum	—	Kochi

38. Consider the following statements:

1. Balaghat is known for its diamond mines.

2. Majhgawan is known for its manganese deposits?

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2007

Ans. (d)

Balaghat, Jabalpur, Jhabua and Chhindwara districts of Madhya Pradesh are an important region for manganese-ore production in India. In Balaghat, the main areas of mining are Katgaria, Langur, Verwali, Netra, Torodi, Batjari, Salwa and Chikpara. Diamond is found in the Majhgawan area in Panna district of Madhya Pradesh. The reserve of diamond in this area is about one million carats. So both the statements (1) and (2) are not correct. Hence, option (d) is the correct answer.

39. Granite slabs and slates are made at –

- (a) Lalitpur (b) Jhansi
(c) Chunar (d) Churk

U.P.P.C.S. (Pre) 2002

Ans. (a)

Granite is found in three districts of Uttar Pradesh - Lalitpur, Mahoba and Banda. Granite slates are widely manufactured in Lalitpur.

40. Where are diamond mines in India?

- (a) Karnataka (b) Uttar Pradesh
(c) Madhya Pradesh (d) Tamil Nadu

U.P.P.C.S. (Mains) 2014

Ans. (c)

Panna is a beautiful town located in Madhya Pradesh. It is the biggest centre of diamond mining in India. Here, diamond mines are located in Panna & Satna districts of Madhya Pradesh.

41. In which one of the following districts, have large reserves of diamond-bearing kimberlite been discovered in the recent past?

- (a) Hoshangabad (b) Raipur
(c) Sambalpur (d) Warangal

I.A.S. (Pre) 2007

Ans. (b)

Large reserves of diamond-bearing kimberlite tracts have been discovered in Payalikhhand and Behradin in Raipur district and Tokapal in Bastar.

42. Which of the following metals are found in Sonbhadra district? Select the correct answer by using the codes given below–

1. Uranium 2. Andalusite
3. Pyrite 4. Dolomite

Code :

- (a) 1, 2, 3 (b) 2, 3, 4
(c) 1, 3, 4 (d) 1, 2, 3 & 4

U.P.U.D.A./L.D.A. (Pre) 2001

Ans. (b)

Uranium is not found in Sonbhadra district. Fireclay, limestone, china clay, dolomite and potash are found in Sonbhadra district. Andalusite and calcite are obtained from Mirzapur. Uranium is found in Lalitpur district.

43. Which of the following substances are found in the beach sands of many parts of Kerala?

1. Ilmenite 2. Zircon
3. Sillimanite 4. Tungsten

Select the correct answer using the codes given below:

- (a) 1, 2, 3 and 4 (b) 1, 2 and 3
(c) 3 and 4 only (d) 1 and 2

I.A.S. (Pre) 2006

Ans. (b)

The heavy mineral sand deposits in Kerala contain an assemblage of Ilmenite, Rutile, Leucoxene, Monazite, Zircon and Sillimanite. The state possesses one of the world class deposits of mineral sand in the coastal tracts between Neendakara and Kayamkulam. Hence (b) is the correct answer.

44. Ilmenite and rutile, abundantly available in certain coastal tracts of India, are rich sources of which one of the following?

- (a) Aluminium (b) Copper
(c) Iron (d) Titanium

I.A.S. (Pre) 2023

Ans. (d)

India has abundant heavy mineral resources, most of which are found along the country's coast and also in inland placers. The seven minerals that constitute heavy mineral sands include rutile, zircon, sillimanite, garnet, ilmenite, monazite and leucoxene (brown ilmenite). The main sources of titanium are rutile and ilmenite.

45. Which State in India is the leading producer of thorium?

- (a) Kerala (b) Bihar
(c) Jharkhand (d) Madhya Pradesh
(e) None of the above / More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (a)

Kerala is the largest producer of thorium in India. Production of monazite is the main source of thorium. Indian Rare Earths Limited produces thorium by processing monazite at its plant located at Udyagmandal (Aluva) in Ernakulam district.

46. Which of the following occurs at coastal beaches of Kerala?

- (a) Beryllium (b) Uranium
(c) Thorium (d) Monazite

M.P.P.C.S. (Pre) 1992

Ans. (*)

Monazite minerals are found on Kerala's coast. Thorium and uranium are also found in monazite minerals.

47. Which of the following minerals occurs in Kerala abundantly?

- (a) Tin (b) Manganese
(c) Mica (d) Monazite

U.P.U.D.A./L.D.A. (Pre) 2013

Ans. (d)

See the explanation of above question.

48. In India, Monazite is found in –

- (a) Uttar Pradesh (b) Himachal Pradesh
(c) Tamil Nadu (d) Kerala

U.P.P.C.S. (Mains) 2002

Ans. (*)

Monazite in India is found in the states of Odisha, Andhra Pradesh, Tamil Nadu, Kerala, West Bengal, Maharashtra and Gujarat.

49. Consider the following statements :

- 1. India does not have any deposits of Thorium.**
2. Kerala's monazite sands contain Uranium.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2009

Ans. (b)

Monazite is an important ore for thorium, lanthanum, uranium and cerium. It is often found in coastal areas of Kerala in large quantities. India, Madagascar, and South Africa have large deposits of monazite sands. The deposits in India are particularly rich in monazite. Thus, statement 1 is wrong but 2 is correct.

50. Match List-I with List-II and select the correct answer using the codes given below the lists:

List- I	List- II
(Uranium)	(State Centres)
A. Domiasiat	1. Rajasthan
B. Lambapur	2. Karnataka
C. Rohail	3. Meghalaya
D. Gogi	4. Andhra Pradesh

Code :

	A	B	C	D
(a)	3	4	1	2
(b)	2	1	3	4
(c)	1	3	4	2
(d)	4	2	3	1

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (a)

According to given options, correct matching of Uranium production zones & concerned states is as follows :

Uranium production zones	State
Domiasiat	- Meghalaya
Lambapur	- Andhra Pradesh (Now Telangana)
Rohail	- Rajasthan
Gogi	- Karnataka

51. Jaduguda is famous for –

- (a) Iron ore
- (b) Manganese
- (c) Gold
- (d) Uranium

U.P.P.C.S. (Pre) 2007

Ans. (d)

Jaduguda Mine is a uranium mine in Jaduguda village in the Purbi Singhbhum district of Jharkhand.

52. Which mineral is mined in Turamdih?

- (a) Kyanite
- (b) Asbestos
- (c) Mica
- (d) Uranium

Jharkhand P.C.S. (Pre) 2016

Ans. (d)

Uranium is mined in Turamdih (eastern Singhbhum, Jharkhand). It is approximately 24 km away from Jaduguda. Uranium Corporation of India Limited is responsible for mining here.

53. Which of the following minerals are found in a natural way in the State of Chhattisgarh ?

1. Bauxite
2. Dolomite
3. Iron ore
4. Tin

Select the correct answer using the code given below:

- (a) 1, 2 and 3
- (b) 2, 3 and 4
- (c) 1 and 4
- (d) 1, 2, 3 and 4

I.A.S. (Pre) 2008

Ans. (d)

According to Minerals Yearbook 2022, list of minerals found in a natural way in the State of Chhattisgarh –
 (1) Bauxite - Bastar, Bilaspur, Dantewada, Jashpur, Kanker, Kawardha (Kabirdham), Korba, Raigarh & Sarguja etc. Rajnandgaon and Kabirdham.
 (2) Dolomite - Bilaspur, Janjgir, Bastar, Durg & Raipur etc.
 (3) Iron ore (haematite) - Bastar.
 (4) Tin - Dantewada
 Thus, option (d) is the correct answer.

54. Match the following :

- | | |
|----------------|---------------|
| (A) Palamu | i. Copper |
| (B) Hajaribagh | ii. Manganese |
| (C) Khetri | iii. Mica |
| (D) Keonjhar | iv. Bauxite |

Code :

- | | | | | |
|-----|-------|-------|------|-------|
| | A | B | C | D |
| (a) | (iv) | (iii) | (ii) | (i) |
| (b) | (ii) | (iv) | (i) | (iii) |
| (c) | (iii) | (ii) | (iv) | (i) |
| (d) | (iii) | (iv) | (i) | (ii) |

U.P.P.C.S. (Spl) (Pre) 2008

Ans. (*)

Palamu, famous for bauxite reserves. Hazaribagh is famous for mica reserves. Khetri is famous for copper production. Keonjhar, located in Odisha, is famous for manganese production.

55. The resource in which Chhota Nagpur plateau is prosperous is –

- (a) Skilled labour
- (b) Mineral
- (c) Hydroelectricity
- (d) Fertile soil

R.A.S./R.T.S. (Pre) 2010

Ans. (b)

Chhota Nagpur plateau is prosperous in various types of minerals.

56. Kaimur Plateau is famous for :

- (a) Copper
- (b) Limestone
- (c) Lithium
- (d) Bauxite
- (e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (b)

Kaimur plateau is famous for limestone. Kaimur plateau in Bihar is mainly located in Rohtas and Kaimur districts. This plateau rock is made of limestone and sandstone. There is a cement industry in Banjari on the basis of limestone of this plateau.

57. Match List-I with List-II and select the correct answer using the codes given below the lists:

- | List- I
(Minerals) | List- II
(Location) |
|-----------------------|------------------------|
| A. Coal | 1. Giridih |
| B. Copper | 2. Jayamkondam |
| C. Manganese | 3. Alwar |
| D. Lignite | 4. Dharwar |

Code :

- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 1 | 4 | 3 | 2 |
| (b) | 2 | 3 | 4 | 1 |
| (c) | 1 | 3 | 4 | 2 |
| (d) | 2 | 4 | 3 | 1 |

Jharkhand P.C.S. (Pre) 2011

Ans. (c)

According to given options, correct matching of minerals and concerned locations is as follows :

Minerals	Location
Coal	Giridih (Jharkhand)
Copper	Alwar (Rajasthan)
Manganese	Dharwar (Karnataka)
Lignite	Jayamkondam (Tamil Nadu)

58. Match List-I with List-II and select the correct answer using the code given below the lists:

List- I (Mining Areas)	List- II (Minerals)
A. Gurumahisani	1. Lead
B. Talcher	2. Uranium
C. Jaduguda	3. Iron Ore
D. Zavar	4. Coal

Code :

	A	B	C	D
(a)	3	4	2	1
(b)	3	2	1	4
(c)	2	4	3	1
(d)	1	2	3	4

U.P.P.C.S. (Pre) 2012

Ans. (a)

According to given options, correct matching of mining locations & concerned minerals is as follows :

Mining Areas	Minerals
Gurumahisani	Iron ore
Talcher	Coal
Jaduguda	Uranium
Zavar	Lead

59. With reference to the mineral resources of India, consider the following pairs :

Mineral	90% natural sources in
1. Copper	Jharkhand
2. Nickel	Odisha
3. Tungsten	Kerala

Which of the pairs given above is/are correctly matched?

- (a) 1 and 2 only (b) 2 only
(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2010

Ans. (b)

According to the Indian Minerals Yearbook, 2022, approximately 52.25% of natural source of copper is found in Rajasthan, approximately 93% of the natural source of nickel is found in Odisha and approximately 41.0% of the natural source of tungsten is found in Karnataka. Hence option (b) is correct answer.

60. Match List-I with List-II and select the correct answer from the code given below:

List- I (Mining areas)	List- II (Minerals)
A. Guru Mahisani	1. Zinc
B. Talcher	2. Uranium
C. Jaduguda	3. Iron-Ore
D. Zavar	4. Coal

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	2	4	3	1
(c)	3	4	2	1
(d)	3	2	1	4

U.P. R.O./A.R.O. (Pre) (Re-Exam) 2016

Ans. (c)

Correct match is as follows :

(Mining area)	(Minerals)
Guru Mahisani	Iron-ore
Talcher	Coal
Jaduguda	Uranium
Zavar	Zinc

61. How much salt is produced by Sambhar Lake with respect to the production of salt in India?

- (a) 8.7% (b) 19.2%
(c) 31.3% (d) 41.7%

R.A.S./R.T.S. (Pre) 1992

Ans. (*)

The option given in the question is correct for the then condition of the year 1992. According to data of Indian Minerals Yearbook 2022, India is the 3rd largest salt (common) producing country in the world after China & USA with a production of about 266 lakh tonnes during the year 2021-22. Sea salt constitutes about 80% of the total salt produced in the country. Gujarat (86%) was the leading state followed by Rajasthan (6%) and Tamil Nadu (6%).